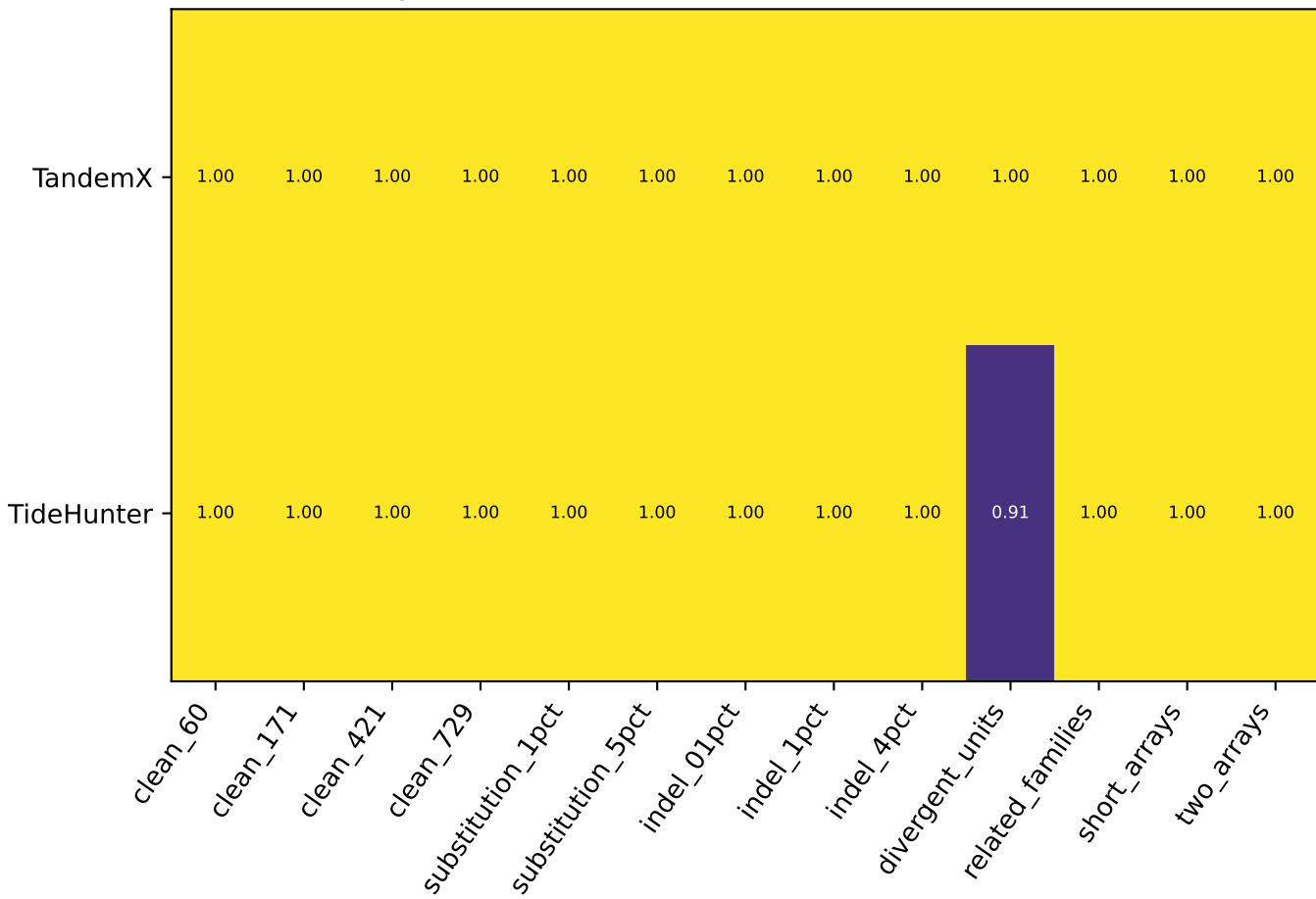
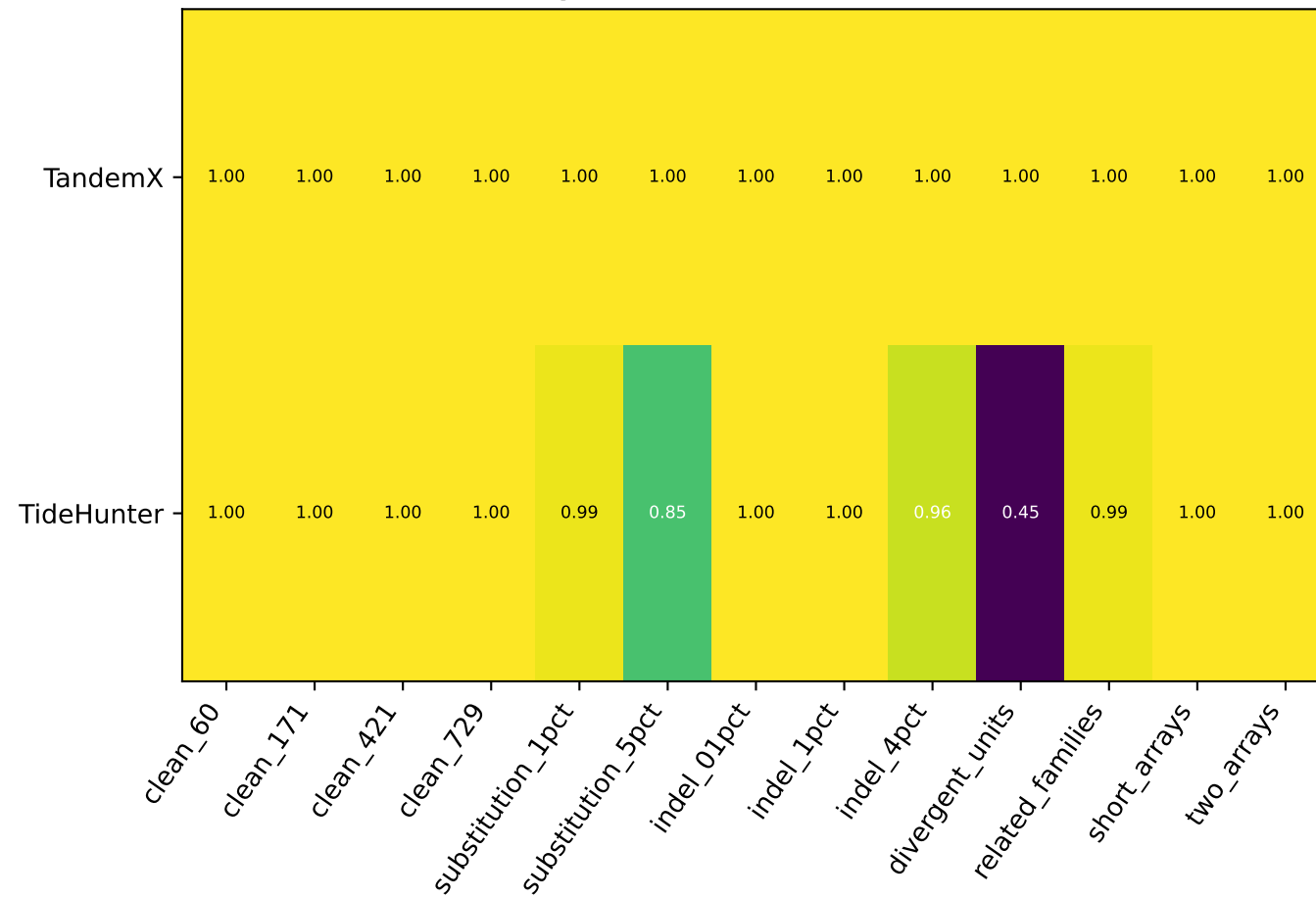


Cascade discovery validation audit: all 14 predeclared gates pass

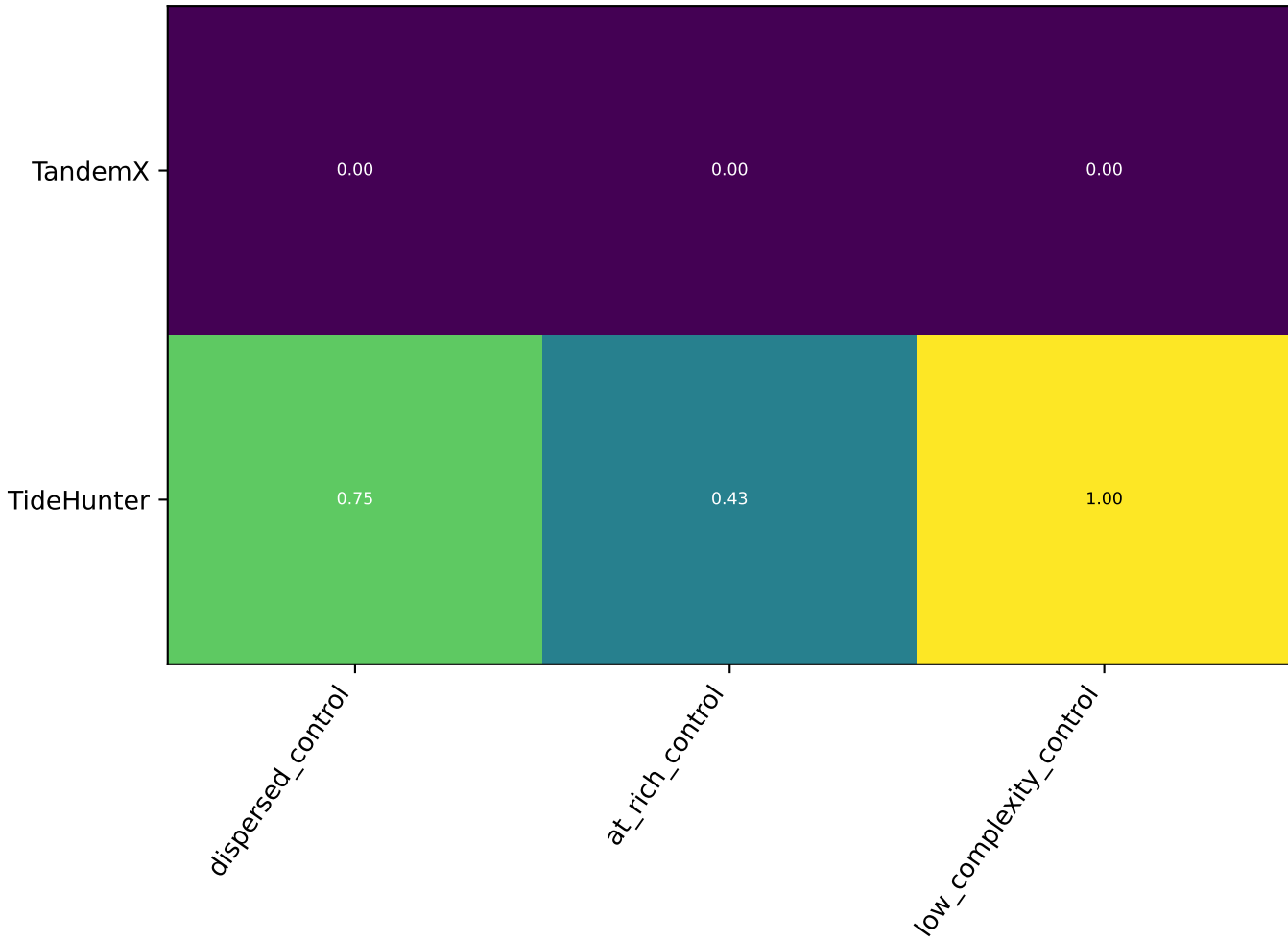
A Mean array recall across validation seeds



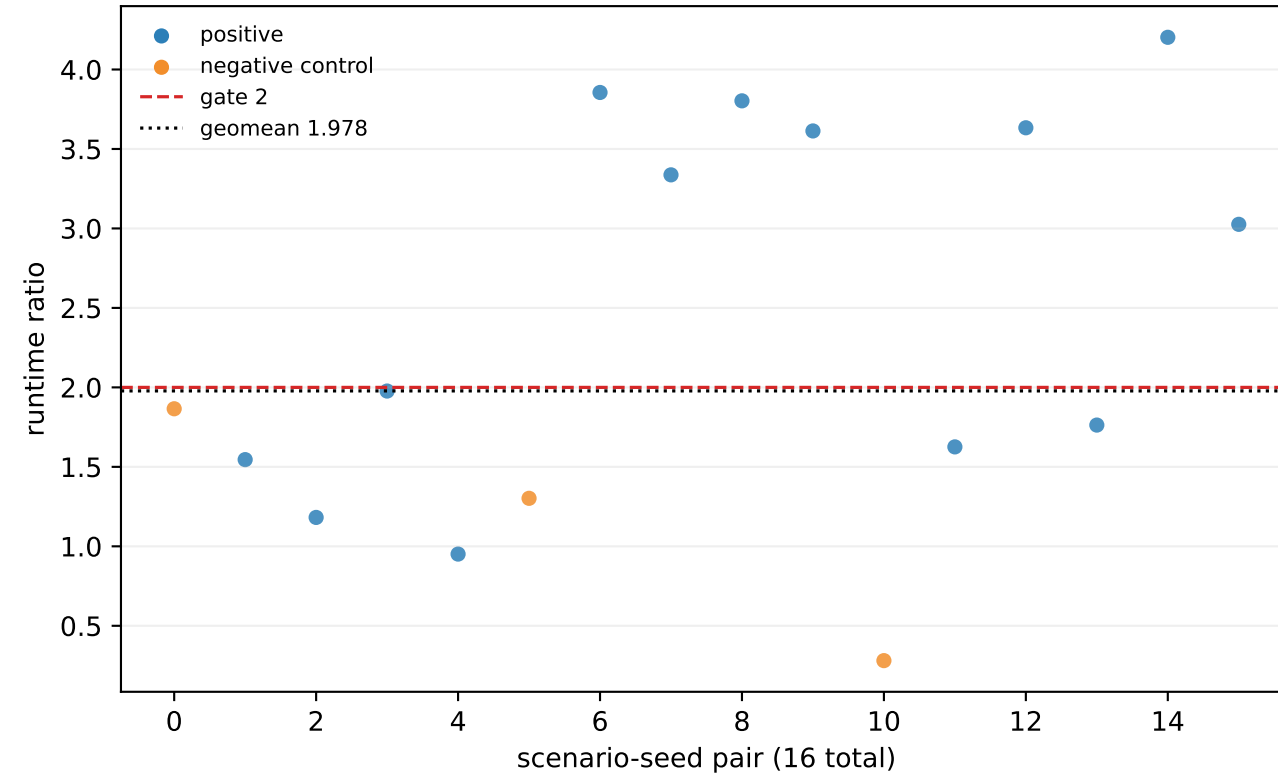
B Mean raw-call array precision



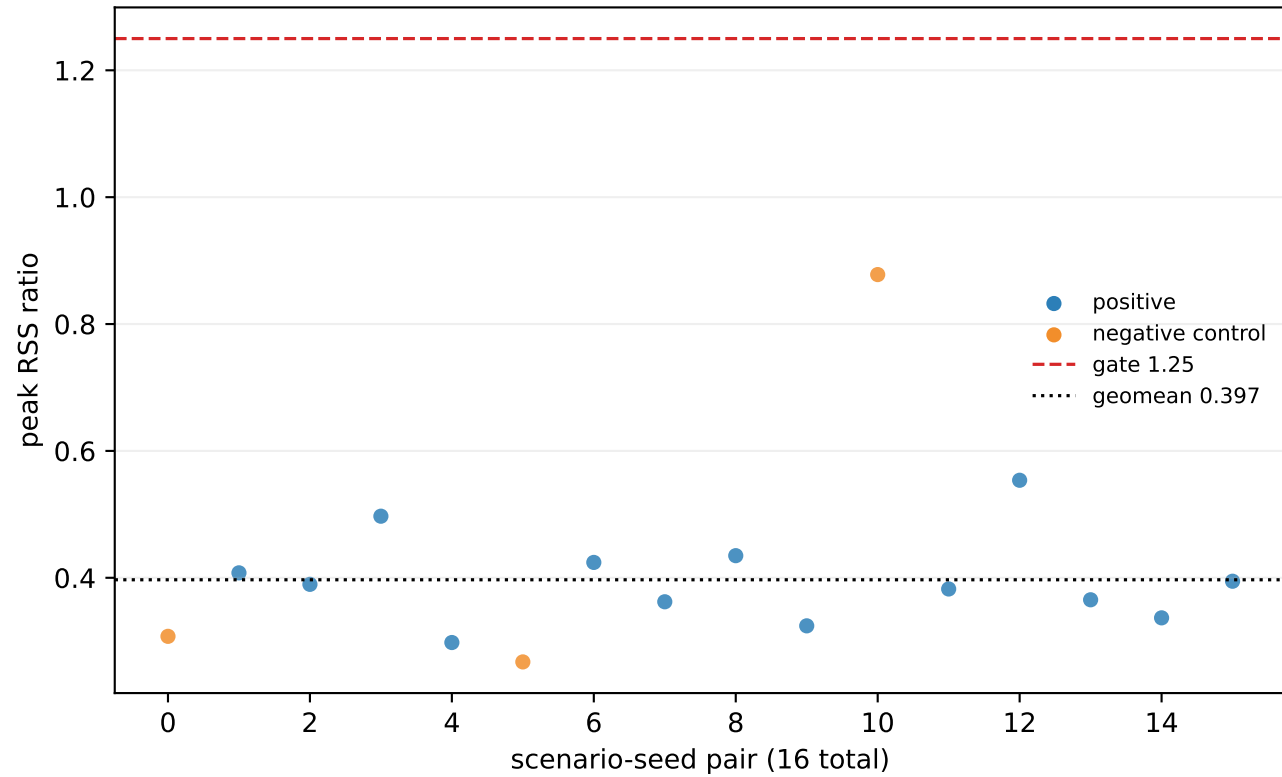
C Negative-read call rate (NA = unavailable)



D Paired TandemX/TideHunter wall-time ratio



E Paired TandemX/TideHunter peak-RSS ratio



F Predeclared evaluation gates

tandemx failed runs	PASS 0 <= 0
all comparator failed runs	PASS 0 <= 0
tandemx deterministic groups fraction	PASS 1 >= 1
minimum positive array recall	PASS 1 >= 0.95
minimum positive array precision	PASS 1 >= 0.95
maximum negative read call rate	PASS 0 <= 0
minimum related family cyclic monomer recall	PASS 1 >= 1
minimum positive base union f1	PASS 0.997 >= 0.995
maximum positive matched boundary mae bp	PASS 2.35 <= 4
paired tidehunter groups fraction	PASS 1 >= 1
worst tidehunter array recall difference	PASS 0 >= -0.02
worst tidehunter array precision difference	PASS 0 >= -0.02
tidehunter runtime geometric mean ratio	PASS 1.98 <= 2
tidehunter peak rss geometric mean ratio	PASS 0.397 <= 1.25